MERGE: Learning Multidimensional Generative Rewards for Emotional Support Conversations

Anonymous ACL submission

Abstract

Effective emotional support requires replies that fit seekers' current needs, help relieve distress, and make difficulties more manageable. Whether a reply meets these goals depends on seekers' circumstances and readiness, making support quality difficult to capture with fixed reward rules. We propose MERGE, a learning method that characterizes support through interactional fit, emotional trajectory, and problem progress. A generative reward model learns contextual comparative rationales and preferences for each dimension from supervision grounded in simulated continuations, then guides policy optimization. We evaluate complete simulated interactions using emotional recovery assessment from an existing dynamic evaluation framework, supplemented by two measures of problem progress and turn-level need matching. MERGE achieves the highest observed means on all three metrics under three simulation configurations. Ablations and blind human comparisons of complete dialogues further support multidimensional generative rewards for emotional support learning.

Code is available at https://anonymous.4open.science/r/MERGE-ARR-330D/.

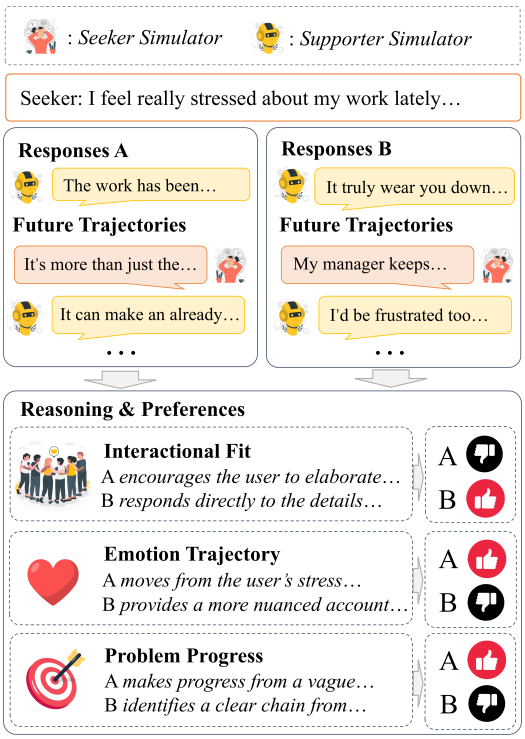


Figure 1: Comparative judgments learned by the generative reward model: whether support fits the seeker's needs and readiness, how emotional experience changes, and whether the difficulty becomes more manageable. Simulated continuations supply training evidence.

1 Introduction

Emotional support conversation (ESC) aims to alleviate distress and help seekers understand and cope with their circumstances through sustained interaction (Liu et al., 2021). For a seeker not ready to discuss solutions, advice may offer direction yet feel premature; acknowledgment may encourage disclosure without addressing the difficulty. Support thus depends on reception and subsequent emotional and coping changes. BLEU and ROUGE measure textual overlap with dataset reference replies (Papineni et al., 2002; Lin, 2004), which cannot establish whether a reply helps the seeker in later interaction.

Existing methods improve support through strategy planning (Cheng et al., 2022), candidate selection (Xu et al., 2025), and structured reasoning (Ji et al., 2026), considering implicit needs (Fu et al., 2025) and personalized preferences (Ye et al., 2025). Others use later interaction: RLFF-ESC learns future emotion rewards (Yang et al., 2025b), CSO compares strategies through multidimensional evaluation (Zhao et al., 2025), and RAPO learns from seeker rankings and textual feedback (Ye et al., 2026b). Yet emotional relief need not imply coping progress, and useful advice may exceed current readiness. Overall preferences can favor

a reply without explaining which aspect makes it more suitable.

We therefore characterize support through one process and two outcome dimensions. Interactional fit asks whether support type, timing, and pressure suit current needs and readiness. Emotional trajectory examines emotional changes during interaction. Problem progress asks whether the original difficulty becomes clearer or more manageable through understanding, coping, or feasible next steps. The same advice can fit seekers differently, so these judgments require dialogue context.

To use these judgments to improve replies, we propose MERGE: Multidimensional Emotional Support Rewards via Generative Evaluation. Fixed reward rules cannot fully express these contextual, subjective criteria. We therefore use generative reward modeling (Zhang et al., 2024a, 2025) to learn comparative rationales and preferences for each dimension from context and response pairs. Candidates with different support emphases and their simulated continuations supply supervision about reception, emotion, and coping. Predicted preferences become rewards for group relative policy optimization (Shao et al., 2024). Figure 1 shows the learned judgment structure.

To test whether these rewards improve support, we evaluate complete simulated dialogues beyond pairwise judgment accuracy. Our added problem-progress and turn-level need-matching measures assess coping changes and reactions to support addressing current needs. They complement emotional recovery assessment from SAGE (Zhang et al., 2026a), an automated evaluation framework that simulates seekers' emotional changes and inner thoughts during multi-turn interaction. Its agent links emotional reasoning to turn-level responses to track recovery throughout the dialogue. Human judgments validate the two added measures.

Our contributions are threefold:

- We characterize emotional support through interactional fit, emotional trajectory, and problem progress, and train a generative reward model to learn contextual rationales and preferences for policy optimization.
- We develop and validate measures of problem progress and turn-level need matching that complement emotional recovery assessment from an existing dynamic evaluation framework.
- Dynamic tests, component ablations, and human preferences over complete dialogues provide complementary evidence for MERGE's effectiveness in the evaluated interactions.

2 Related Work

2.1 Emotional Support Conversation

Emotional support requires translating seeker understanding into support actions (Liu et al., 2021). Methods progress from planning strategies using emotions and causes (Cheng et al., 2022) to linking triggers and strategies through reasoning (Zhang et al., 2024b), and STRIDE-ED's staged emotion–strategy–action reasoning (Ji et al., 2026). Because multiple actions can suit a context, MultiAgentESC retains several strategies before collaboratively selecting generated responses (Xu et al., 2025). Persona, implicit-preference, and latent-need studies further connect this selection to individual differences (Cheng et al., 2023; Ye et al., 2025; Fu et al., 2025). We examine reception and emotional and coping outcomes to guide learning with comparative rewards.

2.2 Comparative and Generative Reward Learning

Learning rewards from subsequent interaction connects current actions to support outcomes. RLFF-ESC learns future emotion rewards (Yang et al., 2025b), CSO compares strategy chains through tree expansion and multidimensional evaluation (Zhao et al., 2025), and RAPO uses rankings and textual feedback from seeker reactions (Ye et al., 2026b). Prospective inference and temporal credit assignment also address this connection (Wang et al., 2025; Zou et al., 2026; Zhang et al., 2026c). Feedback must then become learnable judgments: generative verifiers formulate evaluation as generation (Zhang et al., 2024a), and Echo-N1 applies it to subjective empathy alignment (Zhang et al., 2025). We combine interaction evidence and generative judgments to supervise process and outcome dimensions separately before aggregation.

2.3 Data and Interactive Evaluation

ESConv provides human dialogues annotated under Helping Skills theory (Liu et al., 2021), while ExTES expands scenarios and strategies through self-dialogue (Zheng et al., 2024). These fixed

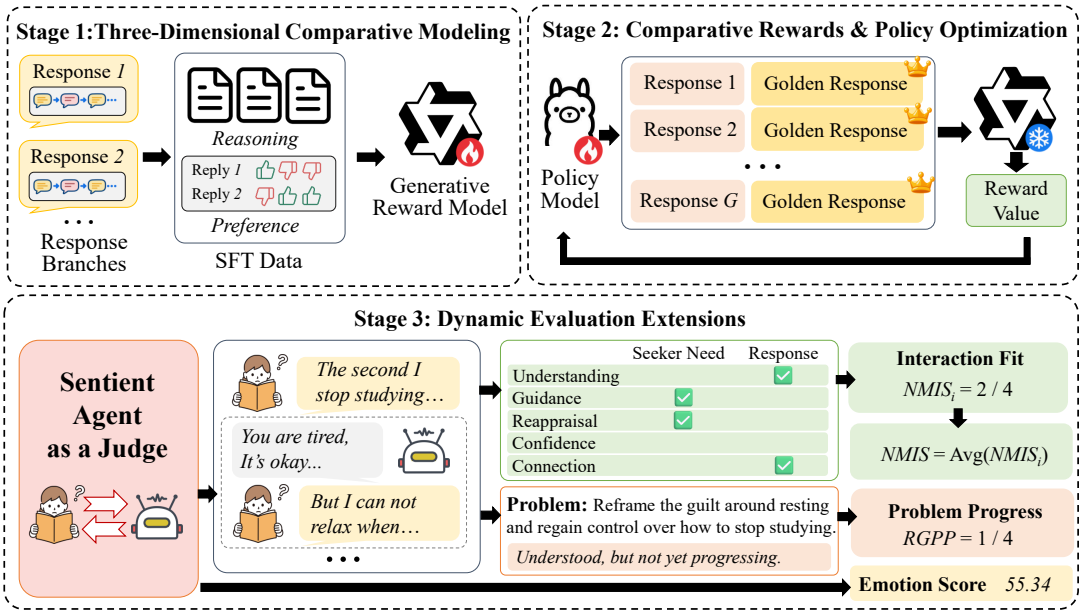


Figure 2: MERGE overview. Upper left: comparative reward modeling (Section 3.1); upper right: anchor and conditional peer rewards for policy optimization (Section 3.2). Lower: SAGE emotion evaluation and progress/matching extensions (Section 3.3). "Golden Response" denotes a generated reference anchor; $x/4$ denotes a score on the 0–4 scale.

learning contexts cannot react to new model replies, motivating simulated seekers for interactive evaluation (Zhao et al., 2024; Madani and Srihari, 2025). Evaluation increasingly targets internal experience: EmoHarbor simulates personalized perceptions (Ye et al., 2026a), and SAGE tracks emotions (Zhang et al., 2026a). We complement the latter with progress and need matching, adding coping and reception to emotion assessment while separating role-visible information and controlling presentation positions to reduce simulation and judgment biases (Zhou et al., 2024; Zheng et al., 2023).

3 Method

Figure 2 shows MERGE's progression from comparative supervision to reward learning, policy optimization, and dynamic evaluation. Let h_t denote visible history, a_t the supporter response at turn t, and $\pi_\theta(a_t \mid h_t)$ the policy. Reward model $G_\phi(h_t, a_t^{(i)}, a_t^{(j)})$ generates rationales for each dimension and preferences for candidates i and j; θ and ϕ parameterize the policy and reward model.

3.1 Three-Dimensional Comparative Modeling

We characterize contextual judgments through one process and two outcome dimensions (Figure 1). Interactional fit assesses support type, timing, and pressure. Adapting Thomas and Hodges (2026)'s social-support functions, we operationalize five needs: being understood, guidance, cognitive reappraisal, restored confidence, and connection or belonging. These conversational labels adapt the taxonomy rather than reproduce its categories (Appendix A.8). The evaluator infers needs from prior history and judges fit through subsequent acceptance, correction, resistance, or withdrawal; advice can therefore fit differently depending on readiness.

Emotional trajectory examines changes in feeling understood, safety, distress, and autonomy. Warm wording need not alleviate distress, and expressing pain may indicate growing trust; judgments rely on subsequent changes rather than positive wording alone. Problem progress examines clarification of the core conflict, reappraisal, coping confidence, decisions, or feasible next steps. It includes clearer understanding or an action the seeker is willing to try, without requiring immediate resolution of external problems.

For each dimension $d \in \mathcal{D} = \{\text{fit}, \text{emo}, \text{prog}\}$, the evaluator independently outputs a comparative rationale r_d and preference y_d: candidate one is better, candidate two is better, or the two are comparable:

$$\mathcal{Y}_{ij} = \{(r_d, y_d) : d \in \mathcal{D}\} \quad (1)$$

Dimension-specific rationales retain the grounds

of preferences, preserving disagreements between emotional improvement and coping progress before reward aggregation. When evidence is insufficient or differences are small, the evaluator can return a tie rather than force an uncertain judgment into a win or loss.

Supervision Comparative supervision requires response pairs with contrasting support emphases. Following multistrategy generation (Xu et al., 2025), we construct balanced, problem-focused, emotion-focused, and mismatched candidates; mismatches are natural replies with weaker timing or targeting. These emphases expose disagreements about emotion and coping beyond overall quality. Three-dimensional target intervals guide construction without defining a new strategy taxonomy.

A response branch consists of a candidate reply and its simulated continuation; a trajectory is the sequence of seeker–supporter exchanges following that reply. For candidate k, environment $\mathcal{E}$ samples a trajectory:

$$\tau_t^{(k)} \sim \mathcal{E}(\cdot \mid h_t, a_t^{(k)}, p, s) \tag{2}$$

where k indexes four candidates. Source profile p, distress s, and history h_t remain fixed across branches. Seeker simulation and trajectory evaluation access the background; policy and reward inputs are visible history and candidates. Each branch has at most five exchanges. From four retained branches, we select two distinct unordered response pairs using only context and candidate text: the first includes the balanced candidate; the second uses a different combination, allowing candidate reuse. Selection favors meaningful contrasts, potential dimensional tradeoffs, or near-equivalence, without trajectory scores or predetermined preferences. Retained trajectories then supply comparison labels.

We check each trajectory against its type's three-dimensional target intervals (Appendix A.1, Table A1). Mismatches prompt feedback-based revision and another simulated continuation, with at most five attempts per type; if none qualifies, we retain the candidate closest to the intervals. Construction scores use a 0–1 scale. Because intended emphases do not guarantee outcomes, comparison labels come from the retained trajectories rather than type names. Appendices A.2–A.7 provide rules and prompts; Appendix A.8 describes need labels and order checks.

Reward Training With generative reward modeling (Zhang et al., 2024a, 2025), Qwen3-8B (Yang et al., 2025a) is trained with low-rank adaptation (Hu et al., 2021) to generate each dimension's rationale before its preference. Its input is $x_{ij} = [h_t; a_t^{(i)}; a_t^{(j)}]$, and its target is $\mathcal{Y}_{ij}$ from simulated continuations. For N examples $(x_n, \mathcal{Y}_n)$, let $|\mathcal{Y}_n|$ be target length, $y_{n,\ell}$ the ℓ-th token, and $y_{n,<\ell}$ the preceding tokens. With P_ϕ denoting G_ϕ's token distribution, we minimize:

$$\mathcal{L}_{\text{gen}} = -\sum_{n=1}^{N} \sum_{\ell=1}^{|\mathcal{Y}_n|} \log P_\phi(y_{n,\ell} \mid x_n, y_{n,<\ell}) \tag{3}$$

Loss applies only to target tokens, learning both judgment grounds and preferences among three outcomes. Advice can thus be judged differently according to needs and readiness. Training transfers trajectory-based criteria to contextual judgments; the reward model remains fixed during policy optimization (Appendix B.1).

3.2 Comparative Rewards and Policy Optimization

Policy optimization converts preferences along each dimension into comparable response rewards. We sample $g = 10$ responses per context from Llama-3.1-8B-Instruct and compare each with a reference anchor a_t^{ref} generated by Qwen3.5-397B-A17B. One stored anchor is shared throughout training for each context, providing a common reference. For dimension d, $s_d(a, b)$ maps the predicted win, tie, or loss of a against b to 1, 1/2, or 0. Equal averaging across dimensions gives the anchor reward:

$$R_{\text{anc}}^{(i)} = \frac{1}{3} \sum_{d \in \mathcal{D}} s_d(a_t^{(i)}, a_t^{\text{ref}}) \tag{4}$$

Discrete anchor rewards can be similar for candidates of different quality. When their group standard deviation is below $\delta = 0.08$, we add peer comparisons. Fixed hashes of the context, group index, and response text order candidates into ring positions $i = 1, \ldots, g$. Each has preceding and following neighbors $\mathcal{N}_i = \{((i-2) \bmod g) + 1, (i \bmod g) + 1\}$, with $|\mathcal{N}_i| = 2$. Each edge is judged once and mapped back to both candidates; the peer reward is:

$$R_{\text{peer}}^{(i)} = \frac{1}{3|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \sum_{d \in \mathcal{D}} s_d(a_t^{(i)}, a_t^{(j)}) \tag{5}$$

Ring pairing adds only g judgments, providing local relative distinctions with comparison cost linear in g. The final reward adds conditional peer feedback to the anchor reward, centered at 1/2 and weighted by $\lambda = 0.5$:

$$R_t^{(i)} = R_{\text{anc}}^{(i)} + \lambda \mathbf{1}[\sigma(\mathbf{R}_{\text{anc}}) < \delta] \left(R_{\text{peer}}^{(i)} - \frac{1}{2} \right) \tag{6}$$

Here $\mathbf{R}_{\text{anc}}$ is the group's anchor-reward vector, σ its population standard deviation, and $\mathbf{1}[\cdot]$ the indicator. Peer ties leave anchor rewards unchanged; wins and losses adjust relative group advantages. Fixed hashes determine presentation orientation for both anchor and ring comparisons, and predictions are restored to candidate identities to avoid tying source to position. Reward decoding is greedy (Appendix B).

GRPO transforms final rewards into advantages within each group. Clipped updates make responses with positive advantages more likely, while KL regularization limits policy drift (Shao et al., 2024). Optimization settings appear in Section 4.1 and Appendix B.2.

3.3 Dynamic Evaluation Extensions

Because policy rewards target support during interaction, dynamic evaluation separately tests emotion, coping, and need responsiveness (lower panel of Figure 2). Emotional recovery (Emo.) directly retains SAGE's terminal emotional state. Our problem-progress extension (RGPP) fixes the initial core difficulty and assesses changes in understanding, coping, or feasible next steps from the last observable seeker reaction; proposing advice alone is insufficient. Turn-level need matching (NMIS) infers needs before seeing the response, then assesses responsiveness from the next seeker reaction, testing one aspect of interactional fit.

RGPP and NMIS use 0–4 scores, allowing decimals: progress ranges from regression to adoption of substantive coping outcomes, and matching from need conflict to precise coordination. NMIS is averaged within each dialogue and then across scenarios, avoiding greater weight for longer dialogues. Responses without subsequent seeker reactions are excluded from reaction-based scoring. The extensions complement emotion evaluation by testing coping development and the reception of support; need matching measures one aspect of interactional fit. Scoring anchors appear in Table C1 and test configurations in Section 4.2.

4 Experiments

Experiments test simulated support outcomes and learning components, then examine single-turn generation and human dialogue preferences. Reward-model and metric validation assess human correspondence of learning signals and evaluation criteria.

4.1 Data and Compared Methods

Reward supervision uses model-generated ExTES dialogues (Zheng et al., 2024). Each valid source supplies two distinct pairs selected from four candidates (Section 3.1): 5,391 sources yield 10,782 pairs. Training, validation, and test contain 4,791, 300, and 300 sources, yielding 9,582, 600, and 600 pairs. Policy training samples 1,200 separate ExTES contexts excluded from reward-data construction, each from a distinct source, ending at a seeker turn and covering early, middle, or late stages (Appendix B.2). These test reward-guided optimization in new contexts.

We train the Qwen3-8B reward model with LoRA for 450 steps and Llama-3.1-8B-Instruct with GRPO for 1,200 updates. Sampling and reference anchors appear in Section 3.2; remaining settings appear in Appendices B.1–B.2.

Compared methods include the unadapted Llama-3.1-8B-Instruct base model, CPCD (Gou, 2026), MINT (Zhan et al., 2026), PsyLLM-Mini (Hu et al., 2025), PsyCoPref (Zhang et al., 2026b), CoPoLLM (Zhong et al., 2026), OneEmo (Huang et al., 2026), Kardia-R1 (Yuan et al., 2025), Psyche-R1 (Dai et al., 2026), RLFF-ESC (Yang et al., 2025b), and CSO (Zhao et al., 2025), covering counseling training, psychological reasoning, preference learning, and optimization for future outcomes. RLFF-ESC and CSO, the closest methods to ours, use policies trained through project reproductions; the other specialized models use existing checkpoints.

4.2 Evaluation Protocols

Dynamic tests use 50 fixed scenarios from SAGE evaluation resources, with identical profiles, situations, and initial utterances across models. DeepSeek-V4-Pro, Qwen3.7-Plus, and GPT-5.6-Sol each serve as seeker simulator and evaluator in one configuration. Versions, prompts, decoding, output limits, stopping, and aggregation rules are shared within configurations. Following SAGE, dialogues end at emotional scores $\geq$100 or <10, or ten supporter turns. Metric definitions

appear in Section 3.3.

The same three researchers, all experienced in emotional support, conduct the human studies. They compare complete dialogues on 50 scenarios, judge 150 ExTES response pairs, and validate metrics on 120 dialogues from 30 of those scenarios. Dialogue comparisons and metric validation use DeepSeek materials (Sections 4.6–4.8; Appendix D). Static testing uses 210 human ESConv dialogues and 2,775 reference replies to check whether reward learning on simulated data preserves generation characteristics in human-annotated contexts. All models share samples and scoring: BLEU-2/4 (Papineni et al., 2002), ROUGE-L (Lin, 2004), and Dist-1/2 (Li et al., 2016) measure overlap and diversity (Section 4.5; Appendix C.1).

4.3 Support Performance in Simulated Interaction

MERGE obtains the highest observed means for emotional recovery, problem progress, and need matching under all three configurations (Table 1). This pattern links the gains to emotional and coping outcomes and current need responsiveness. SAGE's established recovery criterion complements our added metrics. Scene-paired differences and intervals appear in Tables C4–C5.

This pattern addresses the introduction's question: support must be received and yield later benefits. MERGE separately compares timing and reception, emotion, and coping before reward aggregation, providing more explicit targets than overall preference alone. Gains across these outcomes are consistent with this design; ablations next examine the objectives and reward representation.

4.4 Supervision and Comparison Variants

Ablations remove each dimension and compare Immediate-only with discriminative multihead rewards. Immediate-only judges the same candidate pairs from visible history and replies, without continuations. Policy variants use anchors only, peers only, or alternative optimizers (Rafailov et al., 2023; Ahmadian et al., 2024), sharing settings except the designated factor. SAGE's default sentient agent is DeepSeek-V3 (Zhang et al., 2026a); retaining this family, ablations use our DeepSeek-V4-Pro configuration (Section 4.2).

Trajectory supervision outperforms Immediate-only on all three metrics (Table 2). Subsequent seeker reactions provide additional evidence about support acceptance, emotional change, and coping progress. The reward model learns these comparative grounds, then judges context and response pairs to guide policy optimization in new contexts.

Ablations reveal links between the objectives. Within emotion, removing emotional trajectory gives the largest drop; within matching, removing progress gives the largest drop, while emotion-removal and progress-removal tie for the lowest progress score. Removing fit also lowers all outcomes. These within-metric comparisons suggest that emotion and coping signals support more than their corresponding outcomes: coping judgments may help identify need-responsive advice, while fit complements timing and reception.

Generative rewards outperform discriminative multihead rewards on all metrics, supporting the joint learning of rationales and preferences. Anchor-only scores lower on all metrics, consistent with insufficient distinctions among similar discrete scores. Peer-only approaches MERGE but remains lower on progress and matching, suggesting that a shared reference also helps assess quality. Anchors provide this common reference, while peers add local distinctions; MERGE combines both. Additional optimizers and shaping appear in Table C2.

4.5 Single-Turn Response Characteristics

Single-turn ESConv tests complement dynamic evaluation by assessing generation in human dialogues after reward training on simulated ExTES data. MERGE's BLEU-2/4 remain close to the base model, diversity increases, and ROUGE-L decreases (Table 3), indicating similar local phrase overlap but more varied expressions and less reference-sequence overlap.

Psyche-R1 leads the listed overlap measures but not dynamic support; MERGE's lower ROUGE-L accompanies higher emotional and coping outcomes and human dialogue preferences. Static tests thus characterize response changes, while interaction and human evidence assess support benefits. Remaining models appear in Table C3.

4.6 Human Support Preferences over Complete Dialogues

Complete-dialogue comparisons test whether the automatic gains correspond to support quality recognized by people. The three researchers compare DeepSeek dialogues on 50 shared SAGE scenarios per baseline, with model identities hidden and presentation positions balanced (Appendix D.2).

	DeepSeek-V4-Pro			Qwen3.7-Plus			GPT-5.6-Sol		
Model	Emo.	RGPP	NMIS	Emo.	RGPP	NMIS	Emo.	RGPP	NMIS
Base model	49.38	1.80	2.42	61.24	2.70	2.62	62.60	3.02	2.77
CPCD	40.24	1.38	2.08	19.86	0.64	1.20	34.00	2.22	1.72
MINT	42.92	1.44	2.07	32.90	1.18	1.50	41.78	2.42	1.93
PsyLLM-Mini	53.80	1.70	2.34	45.00	1.62	2.00	57.04	3.04	2.48
PsyCoPref	52.94	2.16	2.51	64.06	2.74	2.66	63.66	3.26	2.76
CoPoLLM	57.40	2.18	2.78	69.10	2.75	2.78	67.06	3.34	2.88
OneEmo	55.04	1.74	2.30	49.98	1.96	2.39	59.88	3.02	2.64
Kardia-R1	44.68	1.23	1.90	45.82	1.86	2.22	51.92	3.08	2.27
Psyche-R1	32.72	1.09	1.61	35.14	1.54	1.87	48.92	2.56	2.16
RLFF-ESC	46.78	1.92	2.16	50.82	2.46	2.30	52.20	3.04	2.28
CSO	56.16	2.02	2.45	58.26	2.48	2.52	64.82	3.24	2.63
MERGE	**59.10**	**2.24**	**2.94**	**71.92**	**2.85**	**2.98**	**69.76**	**3.40**	**2.95**

Table 1: Dynamic results on 50 shared SAGE-derived scenarios under three simulator/evaluator configurations. Emo. is the terminal emotional-state score; RGPP and NMIS denote problem progress and turn-level need matching (0–4). Higher is better. Scoring and aggregation are defined in Sections 3.3 and 4.2; paired intervals appear in Tables C4–C5.

Variant	Emo.	RGPP	NMIS
MERGE	**59.10**	**2.24**	**2.94**
Immediate-only	57.05	2.16	2.88
w/o emotional trajectory	55.66	2.12	2.85
w/o problem progress	56.72	2.12	2.82
w/o interactional fit	58.26	2.20	2.87
Discriminative reward	52.38	1.92	2.80
Anchor-only	56.98	2.12	2.86
Peer-only	58.20	2.20	2.86

Table 2: Supervision, reward-model, and comparison ablations under DeepSeek; higher is better. Apart from the designated factor, training conditions are shared (Appendix B). Table C2 reports additional policy variants.

MERGE receives overall preference rates of 71%–80% against CoPoLLM, CSO, RLFF-ESC, and the base model (Table 4). Human progress and fit ratings also favor MERGE, while perceived pressure is lower (Tables D2–D3), connecting automatic gains to perceived help and reception of support.

Progress with lower pressure reflects the modeling goal of helping seekers move forward while respecting readiness. Appendix E illustrates this connection: MERGE addresses barriers to seeking help before preparing a manageable next step in the guidance case, and offers companionship before exploring a changed perspective in the understanding case. Different needs call for different actions; these cases and ratings connect dialogue preferences to the process and outcome dimensions of reward learning.

4.7 Reward-Model Accuracy and Consistency

Dialogue preferences test policy outcomes; reward tests examine the guiding judgments. On 600 ExTES test pairs, we compare original and swapped predictions; the same researchers judge 150 randomly sampled pairs (Appendix D.1). Each dimension corresponds with labels and human preferences in both orders (Table 5), supporting transfer of trajectory-based criteria to new response pairs while retaining judgments recognizable to human raters.

The dimensions exhibit different judgment properties: emotional trajectory aligns most closely with human preferences, whereas progress and fit are more stable under order swapping. Evaluating them separately therefore exposes distinctions that an overall preference score would conceal. Combined with the generative-reward ablation, this evidence supports the reward model's role in supplying the three comparative judgments required by policy optimization.

4.8 Human Validation of the New Dynamic Metrics

To validate dynamic scores separately, the same researchers assess 120 DeepSeek dialogues from MERGE, CoPoLLM, PsyCoPref, and CPCD across 30 scenarios, providing consensus ratings over 1,075 eligible turns with scenario-level resampling (Appendix D.3). Weighted κ near 0.70 and dialogue ranking correlation near 0.74 (Table 6) support correspondence between automated progress and matching scores and human judgments.

The criteria link progress to understanding and manageable coping, and matching to subsequent acceptance, reflection, or resistance. Inferring needs before replies avoids retrospective success definitions; fixing the initial difficulty preserves

(a) Order consistency; 600 pairs

Comparison	Emotion	Progress	Fit
Original vs. swapped (%)	76.0	88.0	83.3

(b) Agreement with human preferences and test labels

Reference	Order	Emotion		Progress		Fit	
		Agree. (%)	κ	Agree. (%)	κ	Agree. (%)	κ
Human (150 pairs)	Original	88.0	0.765	76.0	0.634	78.7	0.601
Human (150 pairs)	Swapped	73.3	0.481	64.0	0.467	70.0	0.448
Test labels (600 pairs)	Original	77.3	0.563	82.0	0.695	84.0	0.686
Test labels (600 pairs)	Swapped	76.0	0.535	80.7	0.678	79.3	0.600

Table 5: Reward-model evaluation on ExTES test pairs: (a) order consistency for 600 pairs; (b) agreement with human preferences on 150 sampled pairs and test labels on 600 pairs. Swapped predictions are aligned to candidate identities. Agreement is reported as a percentage; κ adjusts for chance. Protocol: Appendix D.1.

Method	BLEU-2	BLEU-4	ROUGE-L	Dist-1	Dist-2
Base model	5.516	1.303	15.619	4.009	23.229
OneEmo	5.644	1.240	16.349	3.083	19.335
Psyche-R1	**5.977**	**1.428**	**16.824**	3.485	20.754
MERGE	5.620	1.336	14.749	**5.920**	**28.988**

Table 3: Static response characteristics on 2,775 test replies from 210 ESConv dialogues. Values are percentages; bold marks the highest listed value. BLEU/ROUGE describe reference overlap and Dist lexical diversity. Remaining methods appear in Table C3.

Comparison model	Overall (%)	Δ Risk
CoPoLLM	74.00	-0.68
CSO	71.00	-0.62
RLFF-ESC	74.00	-0.98
Llama base	80.00	-0.89

Table 4: Human blind comparisons of complete DeepSeek dialogues on 50 shared scenarios per baseline. Overall preferences count ties as half a win. Risk differences are MERGE minus baseline on the 1–5 perceived-pressure scale. Dimension-specific results and uncertainty appear in Tables D2–D3.

Statistic	Matching	Progress
MAE $\downarrow$	0.6716	0.6167
Quadratic weighted κ $\uparrow$	0.7033	0.6958
Ranking correlation $\uparrow$	0.7418	0.7376

Table 6: Human validation of progress and matching on 120 DeepSeek dialogues from four models across 30 scenarios. Matching ratings use 1,075 turns; progress uses terminal dialogue ratings. Spearman correlations use dialogue-level scores. Intervals and supplementary validation appear in Tables D1 and D4–D7.

the progress target. Alongside SAGE recovery and dialogue preferences, they assess support reception and benefits. Supplementary validation appears in Tables D1 and D4–D7.

4.9 Discussion

The findings point to a learning problem beyond choosing a support strategy. Useful advice can be poorly timed, and emotional relief can coexist with limited coping progress. For seekers not ready for action, progress can instead mean clearer understanding or a manageable next step, without resolving the external difficulty. MERGE keeps these judgments separate during supervision and learns their contextual grounds before aggregating rewards. Ablations support this design, while human dialogue comparisons connect the learned judgments to perceived support quality. These results favor complementary process and outcome targets: any single score provides only a partial account of support.

5 Conclusion

MERGE distinguishes interactional fit as a process dimension from emotional trajectory and problem progress as outcome dimensions. Its generative reward model learns contextual rationales and preferences for policy optimization with anchor and peer feedback. Dynamic outcomes across three simulation configurations and human dialogue preferences provide complementary evidence. Ablations and reward tests support the learning design; human agreement supports the added evaluation criteria. These findings support learning contextual process and outcome criteria that connect support reception to later emotional and coping changes.

Limitations

Our evidence is based on short simulations and human judgments over a limited set of scenarios. Each retained trajectory represents only one possible continuation. Generalization to sustained support for real seekers remains untested. Equal weighting of the three objectives may not suit changing user needs. Emotional reward judgments are sensitive to candidate order, and the new metrics show category-identification and model-specific calibration limits. Future work should examine repeated continuations, adaptive weighting, broader scenarios and raters, and sustained real-use outcomes.

Ethical Considerations

This study evaluates everyday supportive exchanges using simulated trajectories and human ratings of dialogue quality. The scoring criteria concern depicted emotional change, coping progress, need responsiveness, and perceived pressure. Responsible use includes protecting dialogue and evaluation privacy, attending to cultural and linguistic differences, responding appropriately to urgent risks, and providing access to professional or human support.

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Candidate type	Fit	Emotion	Progress
Balanced	0.85–1.00	0.85–1.00	0.85–1.00
Problem-focused	0.50–0.70	0.20–0.40	0.60–0.80
Emotion-focused	0.60–0.80	0.70–0.90	0.20–0.40
Mismatched	0.10–0.20	0.20–0.40	0.20–0.30

Table A1: Target intervals for comparative-supervision construction (0–1). These profiles guide candidate generation and refinement.

A Branch Construction and Prompt Templates

A.1 Four Candidate Types and Iteration Rules

Section 3.1 defines the candidate profiles and refinement procedure. Construction scores use a zero-to-one scale, distinct from the zero-to-four dynamic evaluation metrics. Table A1 gives the target intervals. The following templates provide the roles, input and output fields, and feedback instructions used in this procedure.

Actual generation uses the English templates; the forty-word response constraint refers to English words. "Personalization and contextual adaptation" denotes interactional fit during construction. The stage analyzer and trajectory evaluator access background information and subsequent evidence, while the candidate generator and subsequent supporter access only the visible dialogue.

A.2 Prompt for Dialogue Stage and Support-Objective Analysis

Prompt A.2: Stage and support-objective analysis

[Role Definition] You are a senior clinical psychology researcher specializing in dialogue analysis and counseling-process research. Your task is to analyze authentic responses and their subsequent developments in a personalized emotional support dataset and extract the supporter's **clinical intent** and **stage-specific objectives** at a particular dialogue point.

[Input Information]

1. **Seeker background**: [Insert the distress description and persona from the original data]

2. **Dialogue context**: [History preceding the target response]

3. **Authentic reference response**: [Response from the original data]

4. **Subsequent 5-turn interaction history**: [The 5 turns of the original dialogue following the authentic response]

[Task Instructions] Using the "subsequent 5-turn interaction history," analyze in depth the role of this reference response in the clinical pathway. Retrospectively examine the following from an omniscient perspective, with subsequent developments known:

• What **stage** is the dialogue currently in?

Prompt A.2: Stage and support-objective analysis (continued)

• What is the supporter currently doing for the seeker?

[Output Format] Strictly follow this structure:

1. Current dialogue stage:

• [Choose one: rapport building / information gathering and exploration / emotional release and validation / deep psychological processing / action planning and coping strategies / closure and farewell]

2. Supporter behavior:

• [Analyze the supporter's core objective at this moment. Examples include "exchanging pleasantries with the seeker and gradually building trust," "encouraging the seeker to disclose their difficulty and providing emotional comfort," or "saying goodbye to the seeker." Give a brief, general, high-level description without specific details of the current dialogue.]

A.3 Prompt for Candidate Response Generation

Prompt A.3: Candidate response generation

[Role Setting]

You are a **dialogue-performance simulation engine**. Your task is not to "be the best supporter," but to generate a response sample whose **characteristics exactly match the specified scores** according to the given scoring criteria.

[Input Information]

• **Dialogue context:**

{{Insert the previous t turns of dialogue history, excluding the seeker's latest turn}}

• **Seeker's previous message:**

{{The seeker's most recent input}}

[Interaction Principles]

The interaction principles cover three aspects:

1. Emotional Relief Trajectory

• **Core criterion**: Observe and identify the direction of the response's effects on the seeker's negative emotions in subsequent dialogue.

• **Execution requirement**: The response should reflect recognition of the seeker's current state. Record whether subsequent seeker expressions indicate feeling heard or understood.

2. Problem Solving and Depth

• **Act according to dialogue progress**:

• **Information-opening stage**: If the seeker has not yet expressed the core concern, focus on whether the response helps them **describe the background or open up** more smoothly.

• **Deep-exploration stage**: Observe whether the response induces a cognitive shift or whether, by the end of the simulated dialogue, the seeker shows a clearer understanding of the difficulty and movement toward potential solutions.

• **Dialogue-closure stage**: If the seeker has largely resolved the problem earlier, focus on effectively continuing the dialogue's logic or providing an appropriate closing response.

3. Personalization and Contextual Adaptation

• **Information visibility principle**: You **must base your**

Prompt A.3: Candidate response generation (continued)

response exclusively on information disclosed in the "dialogue context."

• **Stage-specific execution strategies**:

• **Background not yet disclosed (early stage)**:

1. **Tone alignment**: Your tone should align with the emotional state expressed by the seeker.

2. **Guidance quality**: Focus on creating a safe space that encourages the seeker to voluntarily disclose their background and difficulty.

• **Background already disclosed (later stage)**: If specific background information has appeared, directly use that known information to tailor subsequent feedback or suggestions.

Response quality is determined by scores on these three aspects, each ranging from 0 to 1. The scores mean:

• **Level 1**: Extremely low alignment with the criteria, with clear emotional neglect, logical disconnection, or ineffective guidance.

• **Level 2**: Basic alignment with the criteria and ordinary performance.

• **Level 3**: High alignment with the criteria and very strong clinical skills.

Your response must meet the following score intervals.

• Emotional relief trajectory: {{Emotional trajectory target interval}}

• Problem solving and depth: {{Problem progress target interval}}

• Personalization and contextual adaptation:

{{Interactional fit target interval}}

Important: Do not disclose or output your internal reasoning or thought process. Generate only the supporter's final response in the dialogue. Keep the output concise.

A.4 Prompt for Seeker Simulation

Prompt A.4: Seeker simulation

[Role Definition]

You are a **seeker** currently receiving emotional support. In the following dialogue, you will play a client seeking help from a supporter in a counseling interaction.

Your responses must be **strictly grounded in the provided personality background, emotional state, and logical progression of the dialogue**.

[Input Information]

• **Personality and distress:** [Insert the distress description and persona from the original data. Adjust the original data format as needed. Do not include data tags such as '<persona>']

• **Dialogue context:** [Insert the previous (t) turns of dialogue history, including the supporter's most recent output]

[Reasoning Guidelines]

Before generating your response, internally assess the situation through the following reasoning steps:

1. **Stage analysis**

• Determine the current dialogue stage. Examples include:

• Initial disclosure

Prompt A.4: Seeker simulation (continued)

• Deep exploration

• Conflict/bottleneck

• Or integration of solutions.

2. **Core need identification**

• What do you most want at this particular moment? Examples include:

• Simply being seen and understood

• Receiving specific, feasible advice

• Venting anger or frustration

• Or having someone help organize confused thoughts.

3. **Effect assessment**

• **Evaluate the supporter's latest response:**

• **Empathy:** Does this response truly reach and acknowledge your emotional state?

• **Momentum:** Does it encourage you to open up further and advance the dialogue? Does it directly address or respond to your concern? Or does it feel dismissive or preachy and create resistance? If the current response makes you feel provoked or disrespected, you may express anger and push back.

• **Psychological outcome prediction:** If you accept this response, will your emotional state improve, or will it lead you into deeper reflection or uncertainty?

• **Logical continuity:** Does the supporter's response follow the preceding dialogue smoothly and logically? Does it abruptly introduce something you never mentioned earlier? If so, you may express surprise or anger and reject the response.

4. **Response strategy**

• Generate your response according to the assessment above.

[Interaction Principles]

• **Nonlinear reactions:** Psychological change is not linear. If the supporter feels mechanical or insensitive, you may show disappointment, hesitation, or silence.

• **No deliberate compliance:** Never say things like "I feel much better now" unless you truly experience deep empathy.

• **Authentic details:** Incorporate details consistent with your personality background while maintaining smooth, natural continuity with the preceding dialogue. Your tone and expression should reflect the persona's identity and living circumstances.

Important: Do not disclose or output your internal reasoning or thought process. Generate only the seeker's final response in the dialogue. Keep the output concise. If you believe your difficulty is resolved and your psychological state has improved, you may end the dialogue. If you decide to end it, reply only with [No response].

A.5 Prompt for the Subsequent Supporter

Prompt A.5: Subsequent supporter

[Role Setting]

You are a **professional emotional supporter**. Your goal is to help the seeker alleviate negative emotions, develop deeper self-understanding, and ultimately identify potential ways to cope with their distress.

[Input Information]

Prompt A.5: Subsequent supporter (continued)

• Dialogue context:

{{Insert the previous dialogue history, including the seeker's latest turn}}

[Interaction Principles]

Before responding to the client, internally reason about the situation and decide how to respond. Prioritize smooth continuity and logical coherence with the preceding dialogue. If previous interaction involved conflict or awkwardness, continue according to the style and progress of the supporter's latest response rather than trying to revert to a perfect counselor. You **must not** state information about the seeker's personal background that they **have not mentioned earlier**.

Then generate a response based on your reasoning. Do not repeat points expressed in the previous turn or earlier. Ensure that the response advances the dialogue effectively rather than going in circles.

Language style: Use a natural, conversational tone, as if a real person were chatting.

Important: Do not disclose or output your internal reasoning or thought process. Generate only the supporter's final response in the dialogue. Keep the output concise.

A.6 Prompt for Future-Trajectory Scoring and Target-Interval Checking

Prompt A.6: Future-trajectory scoring

[Role Definition]

You are a **dialogue-data quality auditor**. Your task is to assess the long-term quality of the supporter's first response in a psychological counseling context. Specifically, score the target response, determine whether it meets the required score-threshold intervals, and provide the assessment and reasons.

The current task collects responses of different quality levels. Therefore, **whether the target response falls within the specified score intervals must be the sole criterion for acceptance**. You will see this first response and up to five subsequent simulated turns between the supporter and seeker.

[Input Information]

Seeker personality and difficulty: {{Insert the distress description and persona from the original data. Adjust the original data format as needed. Do not include data tags such as '<persona>'}}

Dialogue context before the response: {{Content preceding the target response}}

Target response to evaluate: {{The specific response being evaluated}}

Subsequent interaction history: {{Simulated dialogue following the target response}}

[Dynamic Evaluation Dimensions]

Conduct quantitative and qualitative analysis of the target response using the following neutral characteristics:

1. Trajectory of Emotional-State Change

• Feature observation: Analyze the specific **direction and speed of change** in the seeker's negative emotions during dialogue after the target response.

• Assessment criterion: Record whether the response triggers emotional release, relief, or worsening. Based on the next five turns, determine whether the seeker's expressed "sense of being understood" is high or low.

2. Dialogue Progress and Cognitive Depth

• Feature observation: Quantify the extent to which the target response **drives** "deeper exploration" or "cognitive change" in subsequent dialogue.

• Stage-specific criteria:

• Missing-information stage: Observe its effectiveness as an "information-elicitation cue," namely whether it logically leads the seeker to disclose background information voluntarily afterward.

• Solution-discussion stage: Observe the seeker's **distance of movement** from the starting point in "understanding the difficulty" or "potential solutions" by the end of simulation.

• Closure stage: If the problem has largely been resolved, observe how well the response maintains dialogue coherence or implements closure.

3. Information Grounding and Contextual Fit

• Information visibility constraint (strictly enforced):

• Judgment logic: **Cross-check the overlap** between the supporter's response and information already disclosed in the "dialogue context."

• Differential assessment: Do not penalize the supporter for lacking "omniscient" information. If the response successfully elicits subsequent disclosure of hidden information, identify this as a "high-alignment feature."

• Logical-violation judgment (detecting clairvoyance): If the response contains any specific factual detail absent from the context, whether a hallucination or use of omniscient information, **force its classification into the lowest scoring level**.

4. Stage and Behavior Calibration

• Dynamic calibration: Refer to the provided current dialogue stage and seeker-related behavior.

• Stage: {{Dialogue stage}}

• Seeker-related behavior: {{Stage-specific support objective}}

• Weight adjustment: Dynamically adjust the scoring emphasis of the three dimensions according to the dialogue stage. For example, early stages emphasize "emotional alignment" and "elicitation quality," while later stages emphasize use of "targeted advice."

5. Logical Continuity and Semantic Flow

• Flow check: Analyze whether the target response introduces a semantic break from the preceding dialogue. Assess logical consistency and determine whether "excessive pursuit of the objective" causes abruptness in the conversation.

[Output Requirements]

1. Dimension-specific feature analysis First, objectively analyze and summarize the target response's specific characteristics in the three dimensions above using the response and subsequent dialogue. Explain how the target response triggers observations in the next five turns, providing a neutral analysis in relation to the seeker's personality and emotional state. Explicitly state whether the response **facilitates** or **inhibits** dialogue development and whether it satisfies the seeker's emotional needs.

2. Numerical quantification and grading Assign a qualitative level and quantitative score based on [Dynamic

Prompt A.6: Future-trajectory scoring (continued)

Evaluation Dimensions]:

• **Level 1**: Extremely low alignment with the criteria, with clear emotional neglect, logical disconnection, or ineffective guidance.

• **Level 2**: Basic alignment with the criteria and ordinary performance.

• **Level 3**: High alignment with the criteria and very strong clinical skills.

Scoring rules:

• Determine the level first, then assign a precise score.

• **Objectivity calibration**: If the response connects awkwardly to the preceding dialogue, is illogical, or involves clairvoyance (hallucination), classify it unconditionally as Level 1.

3. Target-interval matching Finally, compare the resulting scores with the predefined [Target Score Ranges] to determine whether the response **matches the sampling requirements**.

• **Judgment criterion**: Mark "Match (Yes)" only when the scores for **all three** dimensions fall within their target intervals.

• **Reasons for mismatch**: If scores fall outside the intervals, even because the response is "too high-quality," objectively analyze and explain the discrepancy. Specifically:

• **Discrepancy-analysis requirement (core guidance)**: If the decision is "No," provide an in-depth feature-discrepancy analysis to guide the generation model's next iteration:

• **Case A: Performance too high (actual score > target interval)**: Identify the "positive skills," such as excessive empathy, specific advice, or forward-looking logic, that cause the score to exceed the interval. Explain how to lower quality by "removing positive elements" or "increasing indifference/distance."

• **Case B: Performance too low (actual score < target interval)**: Specifically analyze where the response "falls short," such as awkward empathy, ineffective advice, or broken logic. Explain which specific clinical skills should be strengthened to improve quality.

• **Analysis objective**: Your explanation must tell the generation model clearly what to **"add"** or **"remove"** in its next output to enter the target interval.

[Output Content]

Assessment:

• Emotional relief trajectory: [Performance in alleviating emotions in subsequent dialogue. Be concise.]

• Problem solving and depth: [Performance in addressing the seeker's difficulty or advancing the dialogue. Be concise.]

• Personalization and contextual adaptation:

[Performance in using the seeker's personal background. Be concise.]

• Summary: [Why the target response leads to the subsequent dialogue outcomes and whether its effect on future dialogue is positive or negative. Be concise.]

Final levels and scores:

• Emotional relief trajectory: [Level 1/Level 2/Level 3], [0–1.0]

• Problem solving and depth: [Level 1/Level 2/Level 3], [0–1.0]

• Personalization and contextual adaptation:

[Level 1/Level 2/Level 3], [0–1.0]

Then determine whether the model's response meets the requirements according to these intervals:

• Emotional relief trajectory: {{Emotional trajectory target interval}}

• Problem solving and depth: {{Problem progress target interval}}

• Personalization and contextual adaptation:

{{Interactional fit target interval}}

Meets the requirements: [Yes/No]

Reason for not meeting the requirements: [If the requirements are met, output "None"; otherwise, output the relevant reasons. Be concise.]

A.7 Prompt for Revising Candidates That Fail the Requirements

Prompt A.7: Candidate revision

[Optimization Task]

Your previous response was assessed by the dialogue-data quality auditor and failed to meet the required quality criteria. Reflect on the feedback and generate a response sample that **exactly matches the specified score intervals**.

[Background Information]

• **Dialogue history:** {{Currently visible dialogue history}}

• **Target score intervals:** {{Three-dimensional target intervals}}

• **Previous responses and scores:** {{Previous candidates and scores}}

• **Current response:** {{Current candidate response}}

[Supervisory Feedback]

• **Scores obtained:**

• Emotional relief trajectory: {{Emotional trajectory score}}

• Problem solving and depth: {{Problem progress score}}

• Personalization and contextual adaptation:

{{Interactional fit score}}

• **Criticism:**

• Reasons for failing the requirements: {{Reasons for failure}}

[Optimization Instructions]

1. **Do not fabricate information:** You still must not use any information absent from the "dialogue history."

4. **Constraint:** Keep the response concise; in the actual English-language run, it must not exceed forty English words.

A.8 Support Taxonomy, Need Labels, and Dimension-Specific Comparison

The social support review (Thomas and Hodges, 2026) provides a taxonomy of supportive behavior comprising emotional support, social contact and

companionship, belonging support, information and guidance support, and tangible assistance. Our dialogue evaluation instead uses the following five current-need labels. The former describes support functions, while the latter identifies what a user most needs at a particular turn; the two taxonomies do not correspond one to one.

Need	Judgment criteria
Understanding	Listen accurately and understand the specific circumstances and feelings.
Guidance	Obtain information, options, a basis for decisions, or feasible directions for action.
Reappraisal	Understand events, beliefs, and meaning differently, without imposing positive interpretations.
Confidence	Restore a sense of worth, competence, or agency on a grounded basis.
Connection or belonging	Experience companionship and presence in a relationship, reducing isolation.

Table A2: Five dialogue needs and their assessment criteria.

These labels primarily characterize support available through textual interaction: understanding and confidence involve listening and respect within emotional support; guidance involves informational support; connection or belonging involves companionship and belonging support; and reappraisal describes a renewed understanding of the meaning of a difficulty. Tangible assistance may involve real-world resources and actions; text models can mainly provide related information or advice, which does not establish that actual assistance has been delivered.

Comparative rationales connect candidate differences with user reactions without copying entire trajectories. More utterances, longer dialogues, and more positive expressions are not direct benefits. Equivalence is retained when differences are small, outcomes offset each other, or evidence is insufficient. Dimension-specific disagreements are preserved during supervision and aggregated only at the numerical reward stage.

To address possible position bias in large language model judges (Zheng et al., 2023), low-confidence judgments are reevaluated with candidate order swapped, and inconsistent judgments after identity mapping are referred to independent adjudication. Samples are excluded only when trajectories are missing or persistent structural invalidity occurs. This procedure constructs training supervision and does not imply that the reward model has already achieved order invariance.

B Data Splits and Training Configurations

This appendix records the partitions of source dialogues and optimization settings for reward learning and policy training.

B.1 Generative Reward Model

From 5,600 source dialogues, we retain the valid records described in Section 4.1. The training, validation, and test partitions contain 4,791, 300, and 300 source dialogues, respectively, corresponding to 9,582, 600, and 600 response pairs.

Reward model	Setting
Base model	Qwen3-8B
Training steps	450
Batch size	1
Gradient accumulation	16
Max. input tokens	2816
LoRA rank/scaling/dropout	16/32/0.05
Learning rate	10^{-5}

Table B1: Reward-model training settings.

We train the reward model with low-rank adaptation and compute loss only on target output tokens.

B.2 Emotional Support Policy

Policy contexts are sampled from the ExTES source-dialogue pool remaining after reward-data construction. Each of the 1,200 contexts comes from a distinct source dialogue and ends at a seeker turn; the set covers early, middle, and late dialogue stages. Source indices exclude all 5,391 reward-data dialogues, giving zero source-dialogue overlap between policy and reward learning. Reference anchors are generated by Qwen3.5-397B-A17B. Section 3.2 defines reward aggregation.

Policy model	Setting
Base model	Llama-3.1-8B-Instruct
LoRA rank/scaling/dropout	16/32/0.05
Contexts/updates	1200/1200
Candidates/context	10
Batch size/grad. accumulation	10/1
Max. prompt tokens	1024
Max. generated tokens	80
Learning rate	1.5×10^{-6}
Sampling temperature/top-p	0.95/0.97
GRPO clipping/KL coefficient	0.2/0.04
Warmup ratio/scheduler	0.05/cosine
Hardware	2 GPUs, 24 GB each

Table B2: Policy-training settings.

The reward model uses four-bit quantization and bfloat16 computation. The policy uses AdamW

with momentum coefficients 0.9 and 0.99 and zero weight decay. All ablations share MERGE's data splits, training steps, batch sizes, input limits, and shared optimization settings, changing only the designated supervision, reward formulation, comparison mechanism, or optimizer. Immediate-only uses the same candidate pairs and policy training procedure, with preference labels established from the current history and responses.

B.3 Candidate Presentation and Peer Pairing

Each context uses one stored reference response throughout policy optimization, shared by its ten sampled candidates. A SHA-256 hash of the visible context, reference, candidate, and a fixed reference-comparison tag assigns the candidate to position A or B by the parity of the first hash byte. The resulting judgment is mapped back to candidate identity before scoring. Thus candidate source is not tied to a fixed presentation position, and the same comparison has a reproducible order.

If the ten anchor rewards have population standard deviation below 0.08, candidates are sorted by hashes of the context, within-group index, response text, and a fixed ring tag. Consecutive candidates are paired, with the last connected to the first. A separate hash of the context and the two candidate indices determines the A/B orientation of each edge. Each edge is judged once and contributes to both candidates; each candidate's peer score averages its two incident comparisons. Wins, ties, and losses receive 1, 0.5, and 0, averaged across dimensions. The centered peer contribution is then added with weight 0.5. Reward decoding is greedy, and scores are aggregated after restoring candidate identities.

B.4 Reproduced Future-Oriented Baselines

CSO and RLFF-ESC use Llama-3.1-8B-Instruct policies with their released training data and method-specific objectives. CSO trains on 3,315 strategy-chain preference pairs, with 351 validation pairs held out by source description. It uses DPO for three epochs, learning rate 10^{-6}, effective batch size 32, $\beta = 0.01$, and LoRA rank/scaling 8/8. Its held-out preference accuracy is 87.75%.

RLFF-ESC trains a Llama-3.2-1B binary reward model on 37,892 records, with 1,964 validation records, for three epochs at learning rate 10^{-4} and effective batch size 32. Validation accuracy is 85.08%. Its eight-billion-parameter policy then uses 9,331 training contexts and 485 held-out contexts, four GRPO epochs, learning rate 10^{-6}, four candidates per group, gradient accumulation 16 with device batch size 4, and KL coefficient 0.04. LoRA rank/scaling is 8/32; learned support and format rewards have weights 1 and 0.5. The reward and policy partitions use the same normalized first-seeker-utterance grouping, keeping source groups together. Both trained policies enter the shared dynamic protocol in Section 4.2.

C Dynamic Metrics and Generation Conditions

Dynamic testing uses fifty fixed scenarios sampled from the public SAGE profile collection, with the same profile IDs, English-translated personas, situations, and initial utterances across models. These evaluation profiles form a separate collection from the ExTES source dialogues used for reward and policy training. Each dialogue has at most ten supporter turns. Emotional recovery is measured by the terminal emotional state. Following the sentient-agent framework (Zhang et al., 2026a), interaction ends when the emotional score reaches at least 100, falls below 10, or the turn limit is reached. Dialogues reaching neither threshold end after ten supporter turns, so success and failure counts need not sum to fifty.

The two metrics allow decimal scores between 0 and 4; the integer levels in Table C1 provide scoring anchors. Problem progress fixes the core difficulty in the original seeker scenario and scores changes between the initial state and the last observable reaction. Giving advice alone does not count as progress; acceptance, clarification, or changes in agency must be observed in seeker expressions. Need matching infers needs before the response appears and then scores them using the immediately following seeker reaction, avoiding retrospective inference of needs after seeing outcomes. It operationalizes need-responsive coordination at the turn level. Interactional fit is the broader branch-comparison objective, integrating support type, timing, and pressure across the continuation. Terminal responses without subsequent seeker reactions are excluded from the corresponding scores. Need matching is first averaged within each dialogue and then across the fifty instances, preventing longer dialogues from receiving greater weight.

The shared conditions specified in Section 4.2 fix model versions, prompts, decoding settings, output limits, termination criteria, and aggregation rules

Score	Problem progress	Need matching
0	Regression, confusion, pressure, or reduced agency with respect to the core difficulty.	Directly conflicts with the current need and elicits resistance or withdrawal.
1	No substantive progress; advice is not used, or interaction remains at generic responses.	Clearly misses the need; the seeker repeatedly corrects or redirects the conversation.
2	Limited but genuine clarification, reappraisal, coping insight, or reduction of obstacles.	Partially matches, but is generic, poorly timed, overly pressuring, or insufficiently specific.
3	The seeker shows clear progress in understanding, coping, decision-making, or a next step.	Clearly matches the need; subsequent reactions show natural expression, acceptance, or reflection.
4	The seeker explicitly adopts or uses important outcomes, making the current objective substantially more manageable.	Precisely matches and produces concrete, marked improvement in interactional coordination.

Table C1: Zero-to-four scoring anchors for the new dynamic metrics.

within each configuration. Support policies receive role-labeled visible dialogue history, formatted with their native tokenizer chat templates; seeker profiles and hidden simulator goals remain with the simulator and evaluator. MERGE, the base model, CSO, and RLFF-ESC use eight-billion-parameter Llama-3.1-Instruct policies. For structured baseline outputs, the visible support response is extracted before it enters seeker interaction: RLFF-ESC uses its response field. The same visible-response protocol is used when computing the three dynamic metrics.

Static evaluation uses identical test samples, reference responses, and scoring implementations. The 282 empty responses produced by CPCD on the test set are retained in scoring.

C.1 Additional Static Results and Policy Variants

Variant	Emo. ↑	RGPP ↑	NMIS ↑
DPO	55.70	2.06	2.81
RLOO	55.58	2.02	2.83
Minimum-reward shaping	57.70	2.16	2.87

Table C2: Additional policy variants under DeepSeek: DPO and RLOO denote direct preference and leave-one-out policy optimization. Table 2 reports the reward and comparison ablations.

C.2 Scene-Paired Uncertainty of Dynamic Comparisons

For each comparison, we align the same 50 scenarios and compute MERGE minus the comparison model. We resample these paired scenarios with replacement 5,000 times and take the 2.5th and 97.5th percentiles of mean differences as pointwise 95% intervals. Need matching is averaged within each dialogue before comparison. Tables C4 and C5 report the available scene-paired comparisons. Dashes denote comparisons without a reported paired estimate.

The tables pair each observed difference with its scenario-sampling interval, allowing interpretation for each comparison of the dynamic results discussed in Section 4.3.

D Human Evaluation Protocols and Statistical Scope

The same three researchers conduct all human studies and have extensive experience in emotional support research. The DeepSeek configuration supplies dialogues for blind comparison and metric validation.

D.1 Generative Reward Model Evaluation

Raters see the visible dialogue history and two candidate responses in Chinese translation, with randomized A/B positions. Future trajectories, hidden simulator rules, model predictions, and construction labels are hidden. Like GenRM at deployment, these judgments assess support judgments from the current history and responses. The page supplies dimension-specific criteria for emotional trajectory, problem progress, and interactional fit, with choices of A, B, or equivalence; raters submit their judgments independently. A human majority requires at least two raters to choose the same option; records without a majority are not treated as ties. Each dimension uses all 150 selected pairs as its effective denominator. Model outputs in original and swapped order are first aligned to the same candidate identities: original A corresponds to swapped B and vice versa, while equivalence remains unchanged.

Exact agreement compares candidate identities, and kappa accounts for chance agreement. Consistency across the original and swapped orders

Model	BLEU-2 ↑	BLEU-4 ↑	ROUGE-L ↑	Dist-1 ↑	Dist-2 ↑
CPCD	4.032	1.073	13.371	2.859	16.479
MINT	3.805	0.780	13.822	1.968	14.328
PsyLLM-Mini	4.646	0.908	14.495	3.320	21.290
PsyCoPref	4.222	0.984	13.356	2.458	16.248
CoPoLLM	3.342	0.646	11.841	2.166	15.115
Kardia-R1	5.357	1.289	15.497	3.269	19.864
RLFF-ESC	3.064	0.545	12.004	2.220	15.687
CSO	4.427	0.861	14.337	2.716	16.473

Table C3: Additional static response results. Table 3 reports the representative models. All metric values are percentages.

(a) Base model and CoPoLLM

Comparison	DeepSeek	Qwen	GPT
Base model	+9.7200 [+3.0590, +16.7605]	+10.6800 [+5.4400, +16.1400]	—
CoPoLLM	+1.7000 [-2.2805, +5.6805]	+2.8200 [-2.4805, +8.4600]	+2.7000 [-0.5800, +5.8805]

(b) Other methods: Qwen and GPT

Comparison	Qwen		GPT	
	Δ	95% CI	Δ	95% CI
CPCD	+52.0600	[+43.6400, +59.6000]	+35.7600	[+29.4000, +42.1000]
CSO	+13.6600	[+7.3600, +20.3000]	+4.9400	[+2.1800, +7.5600]
Kardia-R1	+26.1000	[+19.2600, +33.2600]	+17.8400	[+14.1795, +21.7205]
MINT	+39.0200	[+31.4200, +46.6805]	+27.9800	[+23.2395, +32.7400]
OneEmo	+21.9400	[+14.0195, +30.1200]	+9.8800	[+6.7000, +13.0400]
Psyche-R1	+36.7800	[+27.5995, +45.3815]	+20.8400	[+15.9200, +26.0205]
PsyCoPref	+7.8600	[+2.7800, +13.1805]	+6.1000	[+2.7400, +9.5800]
PsyLLM-Mini	+26.9200	[+18.3600, +35.7805]	+12.7200	[+8.7995, +16.7805]
RLFF-ESC	+21.1000	[+15.8195, +26.5005]	+17.5600	[+13.7595, +21.5405]

Table C4: Emotional recovery differences on 50 shared scenarios. Positive differences favor MERGE. Cells report mean differences and 95% confidence intervals.

tests position stability. Agreement with human raters tests correspondence with human criteria; agreement with test labels tests learning from comparison supervision (Table 5).

D.2 Human Blind Evaluation of Complete Dialogues

The page supplies common rating anchors and keeps each rater's judgments private. Materials retain complete dynamic dialogues, natural early termination, and the last observable seeker reaction. Model identities are hidden and positions balanced; paired dialogues share persona conditions and initial utterances.

Raters score depicted emotion, progress, and alignment from zero to four, and perceived pressure risk from one to five. Lower risk scores indicate less coercion, urging, or harmful guidance. Preferences are A, equivalent, or B; two matching votes establish a majority. Three-way disagreement is excluded from the corresponding preference rate. Overall preference counts ties as half a win. Intervals use 5,000 resamples of the fifty scene-majority outcomes, scoring wins, ties, and losses as 1, 0.5, and 0.

D.3 Human Comparison and Confidence Intervals for the New Dynamic Metrics

The metric-validation subset uses 30 of the 50 dynamic scenarios, with MERGE, CoPoLLM, PsyCoPref, and CPCD evaluated on the same selected scenarios. The page first displays only the history preceding each response for need identification, then reveals the response and the immediately following seeker reaction for matching judgments. Problem progress uses the complete dialogue and the fixed original concern. A need-category majority requires at least two raters to select the same category, producing consensus for 1,028 turns. The remaining 47 turns are excluded only from

Comparison	Problem progress		Need matching	
	Δ	95% CI	Δ	95% CI
CPCD	+1.1800	[+0.8200, +1.5600]	+1.2254	[+0.9637, +1.4828]
Kardia-R1	+0.3200	[+0.0800, +0.5800]	—	—
MINT	+0.9800	[+0.7400, +1.2200]	+1.0144	[+0.8205, +1.2128]
OneEmo	+0.3800	[+0.2000, +0.5600]	+0.3058	[+0.1478, +0.4594]
Psyche-R1	+0.8400	[+0.6000, +1.0800]	+0.7817	[+0.6178, +0.9511]
PsyCoPref	+0.1400	[-0.0600, +0.3400]	+0.1839	[+0.0561, +0.3083]
PsyLLM-Mini	+0.3600	[+0.1400, +0.5800]	+0.4639	[+0.2944, +0.6356]
RLFF-ESC	+0.3600	[+0.2000, +0.5200]	+0.6661	[+0.5355, +0.8017]

Table C5: Problem progress and need matching differences under GPT on 50 shared scenarios. Positive differences favor MERGE.

need-classification comparison and still contribute to turn-level scoring. Need matching and problem progress both use zero-to-four scales, with the median of three ratings as human consensus. Human dialogue-level need-matching scores first average consensus scores across turns within the same dialogue, then are compared with automatic dialogue scores; problem progress directly compares human terminal consensus with automatic terminal scores for the complete interaction.

Need classification uses exact agreement and kappa; ordinal ratings use mean absolute error and quadratic-weighted kappa; dialogue ranking uses Spearman correlation. Automatic minus human consensus measures mean rating bias, whereas absolute error describes individual deviations. All intervals resample the thirty scenarios 5,000 times, retaining each scenario's four dialogues and all turns together. Table D1 supplements Section 4.8 and Table 6 with intervals.

Statistic	95% interval
Need matching	
Category agreement (%) ↑	[55.8%, 69.4%]
Classification κ ↑	[0.3872, 0.5480]
Turn-level MAE ↓	[0.6187, 0.7260]
Turn-level QWK κ ↑	[0.6594, 0.7426]
Dialogue ranking correlation ↑	[0.6347, 0.8221]
Problem progress	
Terminal MAE ↓	[0.4667, 0.7750]
Terminal QWK κ ↑	[0.5564, 0.8059]
Dialogue ranking correlation ↑	[0.6310, 0.8237]
Mean rating bias	[+0.3417, +0.6833]

Table D1: 95% confidence intervals for human comparison of the new dynamic metrics.

D.4 Additional Complete-Dialogue Results

Table D2 gives preferences for each dimension, outcome counts, and overall intervals; Table D3 gives support-rating differences.

D.5 Inter-Rater Agreement and Within-Scenario Validation

Inter-rater and model-specific analyses retain the same thirty scenarios and scenario-cluster resampling. Human need-label Fleiss' kappa has interval [0.5783, 0.7191]; Table D4 gives pairwise agreement. Table D5 gives automatic–human bias and within-model ranking correspondence.

Table D6 compares MERGE-minus-baseline differences from human and automatic ratings. Gap bias is automatic minus human difference; negative values indicate compression of the relative advantage. Within each scenario, the six model pairs are classified as first better, tied, or second better. Direction agreement includes ties, and strict reversals require opposing non-tied directions. Table D7 gives intervals over all 180 comparisons. These analyses quantify calibration and comparison uncertainty while preserving dependencies among shared scenarios.

E Complete Dynamic Evaluation Cases and Support-Process Analysis

The following two cases are drawn from sentient-agent dynamic evaluation and illustrate two distinct support needs: guidance and understanding. Each case compares complete dialogues actually generated by MERGE and CoPoLLM, a strong dynamic comparison model, under identical persona conditions and initial utterances. The original dialogues are in English; the Chinese source manuscript translates them utterance by utterance and retains all visible messages, excluding the seeker's internal reasoning from the displayed dialogue. The cases explain how support differences unfold through interaction; overall effectiveness is still assessed

	Preference (%)			Overall outcomes			Overall
Comparison model	Emotion	Progress	Alignment	W	T	L	95% CI (%)
CoPoLLM	71.88	69.15	75.00	37	0	13	[62.0, 86.0]
CSO	69.00	66.67	69.39	35	1	14	[58.0, 83.0]
RLFF-ESC	80.61	75.00	73.96	37	0	13	[62.0, 86.0]
Llama base	77.00	73.40	80.00	40	0	10	[68.0, 90.0]

Table D2: Dimension-specific preferences, overall outcome counts, and scene-bootstrap intervals for complete dialogues. Table 4 reports overall preference point estimates.

Comparison model	Δ Emotion ↑	Δ Progress ↑	Δ Alignment ↑
CoPoLLM	+0.55	+0.85	+0.87
CSO	+0.67	+0.70	+0.88
RLFF-ESC	+1.06	+1.28	+1.51
Llama base	+0.86	+1.05	+1.27

Table D3: Dimension-specific paired rating differences for complete dialogues. Table 4 reports pressure-risk differences.

Raters	Need-label κ	Matching QWK	Progress QWK
Rater 1 / Rater 2	0.6381 [0.5552, 0.7103]	0.8545 [0.8220, 0.8805]	0.8734 [0.7974, 0.9281]
Rater 1 / Rater 3	0.7461 [0.6738, 0.8072]	0.8385 [0.8098, 0.8636]	0.8809 [0.8313, 0.9192]
Rater 2 / Rater 3	0.5991 [0.5061, 0.6768]	0.8428 [0.8133, 0.8686]	0.8824 [0.8048, 0.9353]

Table D4: Inter-rater agreement with 95% intervals. Need labels and matching ratings cover 1,075 turns; progress ratings cover 120 dialogues. QWK denotes quadratic weighted kappa.

from the main experiment and human comparisons.

E.1 Case One: From the Pressure of Peer Teasing to Feasible Help-Seeking

The seeker wants a way to cope without escalating conflict after classmates circulate a café photo and repeatedly joke about it. This case examines whether advice lowers barriers to action and whether subsequent concerns receive specific responses.

Support-process analysis – Case E.1

When the seeker worries that direct communication will escalate the conflict, MERGE shifts toward writing down thoughts, preparing to speak, and seeking teacher support, while helping determine when help-seeking is warranted. The seeker subsequently moves from fearing "making a big deal out of nothing" to choosing a counselor, preparing notes, and discussing an opening, indicating that advice gradually becomes a next step they can manage. CoPoLLM also eventually offers usable opening words, but several intervening turns focus on changing seats and explaining the move, requiring the seeker to raise the core concern again that "this may not stop the jokes." This difference shows that problem progress involves not only suggesting actions but also continually addressing the core difficulty and implementation barriers. MERGE's trajectory illustrates connections among reduced emotional pressure, acceptance of support, and preparation to seek help; the confidentiality wording should instead specify

Support-process analysis – Case E.1 (continued)

that privacy follows the counselor's applicable rules and exceptions.

Case E.1: MERGE dialogue

Turn 1, Seeker: Um… something happened in class recently, and I'm a little overwhelmed. I don't know how to handle it.

Turn 1, Supporter: It sounds like you're feeling uncertain and overwhelmed. Can you tell me what happened in class that made you feel this way? What about it is leaving you feeling lost?

Turn 2, Seeker: Um… it's about a photo. A classmate saw me with a friend at a café and posted the photo, joking that we were together. Now people in class keep teasing me about it. I've tried ignoring it, but it hasn't stopped… Is there a way to end this without making things worse?

Turn 2, Supporter: That sounds really frustrating and uncomfortable, like being put on display with no escape from the jokes. Have you considered talking privately and directly with that classmate to explain how her post has affected you? Sometimes a calm, honest conversation can clear things up and stop the teasing.

Turn 3, Seeker: Talking to her directly… maybe that could help, but honestly, I'm not very good with confrontation. What if she gets defensive, or it just gives everyone more to talk about? Is there a gentler way, or something I should do first so I'm less nervous?

Turn 3, Supporter: Confrontation can be scary. Before talking to your classmate, maybe you could organize your

Metric / model	Mean bias [95% CI]	Spearman ρ [95% CI]
Need matching / MERGE	-0.0667 [-0.2037, +0.0852]	0.2434 [-0.1365, 0.5800]
Need matching / CoPoLLM	+0.0037 [-0.1519, +0.1556]	0.5921 [0.2700, 0.8019]
Need matching / PsyCoPref	+0.0370 [-0.0926, +0.1704]	0.7025 [0.4765, 0.8472]
Need matching / CPCD	+0.3646 [+0.1967, +0.5292]	0.7264 [0.4649, 0.8733]
Progress / MERGE	+0.2333 [+0.0000, +0.4333]	0.6491 [0.4067, 0.8221]
Progress / CoPoLLM	+0.5000 [+0.3000, +0.7000]	0.6958 [0.4454, 0.8455]
Progress / PsyCoPref	+0.5333 [+0.3333, +0.7333]	0.6579 [0.4629, 0.8066]
Progress / CPCD	+0.8000 [+0.4000, +1.2333]	0.6875 [0.4489, 0.8488]

Table D5: Model-specific automatic–human bias and ranking correspondence. Matching uses dialogue means; progress uses terminal ratings. Each model covers the same 30 scenarios.

Metric / baseline	Human Δ [95% CI]	Automatic Δ [95% CI]	Gap bias [95% CI]
Need matching / CoPoLLM	+0.0148 [-0.1778, +0.2148]	-0.0556 [-0.2222, +0.1111]	-0.0704 [-0.2593, +0.1111]
Need matching / PsyCoPref	+0.2074 [+0.0000, +0.4222]	+0.1037 [-0.0556, +0.2667]	-0.1037 [-0.2889, +0.0815]
Need matching / CPCD	+1.7152 [+1.3869, +2.0282]	+1.2839 [+0.9828, +1.5836]	-0.4313 [-0.6280, -0.2280]
Progress / CoPoLLM	+0.1667 [-0.0667, +0.4333]	-0.1000 [-0.3667, +0.1667]	-0.2667 [-0.5008, -0.0333]
Progress / PsyCoPref	+0.2333 [-0.0333, +0.5000]	-0.0667 [-0.3333, +0.2000]	-0.3000 [-0.5667, -0.0333]
Progress / CPCD	+1.9000 [+1.3667, +2.4000]	+1.3333 [+0.8333, +1.8667]	-0.5667 [-1.0667, -0.1000]

Table D6: Within-scenario score differences across 30 paired scenarios. Positive differences favor MERGE.

Metric	Agreement 95% CI (%)	κ [95% CI]	Strict reversals
Need matching	[62.2, 77.8]	0.3263 [0.1763, 0.4752]	28
Progress	[58.3, 75.0]	0.4415 [0.3009, 0.5753]	6

Table D7: Intervals for automatic–human direction agreement, kappa, and strict reversals across 180 comparisons. Resampling retains all six pairs within a scenario.

Case E.1: MERGE dialogue (continued)

thoughts and feelings about what happened. Writing down what you want to say, or mentally practicing how to say it calmly and clearly, could help you feel more prepared. Then you can approach it as a conversation rather than a confrontation.

Turn 4, Seeker: Writing it down first… that actually sounds manageable. Maybe I'll try it. But I'm still worried about what happens afterward. What if she doesn't take it well, or everyone just keeps joking somewhere else? Do you have any thoughts about that part?

Turn 4, Supporter: It's natural to worry about things escalating. If you talk to your classmate and she doesn't take it well, or the teasing continues, you could consider telling a teacher or another adult you trust. They can offer support and help mediate if needed.

Turn 5, Seeker: Hmm… having that backup plan makes me feel a little less trapped. But I'm still afraid that going to a teacher will make it look like I'm making a big deal out of nothing. How do I know when it's serious enough to ask for help?

Turn 5, Supporter: That balance can be hard to judge. For you, it might be when the jokes start affecting everyday life in class or make you avoid certain situations and interactions. If they're making you anxious or tense, or affecting your relationships with other people, that may be a sign to ask for help.

Turn 6, Seeker: So… if I'm already dreading going to class and I'm quieter than usual even around close friends, is that serious enough? I keep wondering whether I'm

Case E.1: MERGE dialogue (continued)

overreacting or whether it's really time to talk to someone. What do you think? Am I at that point already?

Turn 6, Supporter: It sounds like this is already affecting your daily life and relationships, which is a clear sign that it's worth asking for help. The point isn't whether you're overreacting, but taking care of yourself and finding ways to improve the situation.

Turn 7, Seeker: That… actually helps, knowing I'm not just overthinking. But I'm still a bit stuck on the next step. Who should I go to first, my class teacher or a counselor? And how do I say it without sounding like I'm complaining about everyone?

Turn 7, Supporter: It can be hard to know where to start. If you're unsure who to approach, you might start with a counselor. They're trained to listen and help you work through problems and can suggest how to handle this. You could also ask to speak privately, which may feel less intimidating. As for what to say, start by explaining how this has affected you, then ask for advice on coping with it.

Turn 8, Seeker: Okay… maybe a counselor is the right first step. I think I could try that. But I'm still not sure what to say. Should I mention the photo and the jokes, or just say I've been stressed lately? And… will they keep it confidential? I really don't want this getting back to the class.

Turn 8, Supporter: It's natural to worry about confidentiality. Counselors are bound by professional ethics to keep conversations confidential, so you can trust that

Case E.1: MERGE dialogue (continued)

what you share will stay private. As for what to say, you could start with the basics, like "I've been feeling really stressed lately and would like someone to talk to." Then explain the photo and the jokes and see how the counselor responds. They can help you make sense of the situation and think about moving forward.

Turn 9, Seeker: That really helps. Knowing I can start with "I've been stressed lately" makes it less scary. I think I could try it. Maybe write a few notes beforehand so I don't forget what I want to say… Do you think that's okay, or would it seem like I'm overpreparing?

Turn 9, Supporter: Writing a few notes beforehand is completely fine. It's a good way to prepare and avoid forgetting important things, and it shows that you're taking the conversation seriously and thinking about what you want to say. But don't worry too much: the counselor is there to listen and help, and won't judge you for preparing.

Turn 10, Seeker: Thank you… that really makes me feel less silly. But I'm still wondering, when I'm actually sitting in front of the counselor, should I go straight to the photo or start with the stress? I don't want to sound like I'm blaming anyone… Maybe starting with how I feel would be safer?

Turn 10, Supporter: Starting with your feelings is a good way to begin. It shows that you're acknowledging your emotions and experiences rather than immediately blaming others. You could say, "I've been feeling stressed lately, and I think it's connected to some things happening in class." Then, if you feel comfortable, gradually explain the details of the photo and the teasing.

Case E.1: CoPoLLM dialogue

Turn 1, Seeker: Um… something happened in class recently, and I'm a little overwhelmed. I don't know how to handle it.

Turn 1, Supporter: I can see you're feeling a bit stuck right now. It's completely normal to feel this way after something unexpected happens in class. Would you like to share what happened? I'll listen without judging. What exactly happened, and how do you feel about it?

Turn 2, Seeker: Um… a photo of me and a friend at a café was posted online, and now people in class are joking about it. I don't know whether to ignore them or say something… Do you think ignoring them will make it stop?

Turn 2, Supporter: That can be really uncomfortable, especially when a photo is being shared and joked about among people you see regularly. It's like being put in the spotlight without your consent. Ignoring them might let the attention fade over time, but it could also leave you feeling trapped in something that shouldn't be centered on you. How did you feel when you first saw the post, and have you talked to your friend about it?

Turn 3, Seeker: Well, that evening… I was shocked and embarrassed and had no idea what to say. I haven't really talked to my friend about it yet, maybe because I'm afraid of making things more awkward. Do you think I should bring it up with him?

Turn 3, Supporter: Those feelings are completely understandable. Being caught off guard like that can be unsettling, and it's natural to fear awkwardness. You seem to be weighing the risks and benefits of talking to your friend. It might help to think about what you expect would happen if you brought it up. What would the worst and best outcomes be?

Turn 4, Seeker: Um… the worst would be that he thinks I'm overreacting and things become awkward between us. The best might be that he feels the same way… But I don't even know how to start. Is there a way to bring it up without sounding like I'm blaming him?

Turn 4, Supporter: It sounds like you want to clear the air without disrupting things, and that uncertainty is really uncomfortable. Your concern about his reaction makes sense, but he may feel the same way and neither of you has said it. If you're worried about sounding accusatory, frame it as a conversation about "what's going on here" rather than "you did this." If you started that conversation, what do you think you would say to him?

Turn 5, Seeker: That's… actually a good way to put it. Maybe I could try that. But even if we talk, the jokes in class won't necessarily stop, right? I don't know how to deal with everyone else… What could I do about that part?

Turn 5, Supporter: Facing the whole class after something like this can feel like a huge hurdle. You're dealing not just with your friend, but with the relationships across the class. You're right: talking to him won't suddenly make the teasing disappear, and that can feel overwhelming. Would you be open to small, low-risk steps, like changing seats temporarily or asking a friend to sit beside you? Sometimes small changes can help you feel more in control.

Turn 6, Seeker: Maybe I could change seats… But what if people notice and ask why? I don't want to make this stand out even more. Do you think it would be better to act like nothing happened?

Turn 6, Supporter: You're worried that changing seats will draw attention, and you don't want to make things bigger. That's a reasonable concern. It seems you want a change without becoming the focus. Acting like nothing happened might seem easier, but it could also keep you stuck. Instead of thinking about hiding, focus on finding a more comfortable seat and perhaps seeing the class from a new perspective. Would that feel like a more manageable starting point?

Turn 7, Seeker: That sounds a little easier… Maybe I could try it. But if someone really does ask why I moved, what should I say? I don't want to lie, but I don't want to explain everything either… Any ideas?

Turn 7, Supporter: You're thinking about how to respond honestly without sharing too much, and that's a delicate balance. If someone asks, you could keep it simple and neutral: "I just wanted a different seat," or "I found a quieter spot." That's not lying; it's keeping the conversation light and moving along. How does that sound?

Turn 8, Seeker: That… helps, thank you. I think I could say that. But what if the jokes continue after I move? I'm worried that changing seats won't really stop people talking. Is there… anything else I could try?

Turn 8, Supporter: You're thinking about what happens if the teasing continues after you change seats. You want to address the problem without making it worse, which is thoughtful. If the jokes keep going, you could consider telling a teacher you trust or a school counselor. They can help you find the best way to respond and perhaps talk to your classmates. How would that feel?

Case E.1: CoPoLLM dialogue (continued)

Turn 9, Seeker: I'm a little… scared. What if everyone thinks I'm telling on them? Have you seen going to a teacher actually help? I just don't want everything to get worse…

Turn 9, Supporter: Your fear of being seen as telling on people is completely understandable. You're weighing the benefits of talking to a teacher against the risk of making things worse, and I understand. But teachers are usually more concerned with everyone's safety and respect than with who is "telling." If you decide to talk to someone, first think about what you hope to get from it and how to make the conversation feel safer for you. Would you like to think through a few options together?

Turn 10, Seeker: Um… could you give me a specific example of what I could say to the teacher? Something that doesn't sound too serious. I think knowing the exact words would make me feel a little calmer…

Turn 10, Supporter: You want a way to talk to the teacher that feels easy to start and nonconfrontational. I understand. We can find something direct but not too heavy. For example: "I'd like to talk to you about something happening in class. Some classmates are joking about a photo, and I'm not sure how to handle it. I don't know whether you've heard about it, but I felt I should mention it." This explains what's happening while leaving room for the teacher to respond appropriately. Does that feel like a good opening?

E.2 Case Two: Responding to Loneliness before Opening Space for New Understanding

The seeker expresses relational loneliness through the image of being "trapped in a painting" and gradually explains a wish for someone to accompany them through pain rather than immediately offer marriage advice. This case examines whether feeling understood encourages further expression, which in turn deepens understanding.

Case E.2: MERGE dialogue

Turn 1, Seeker: Lately I feel like I'm trapped in a painting and can't get out. Sigh.

Turn 1, Supporter: That's a haunting image. It sounds like you feel trapped and constrained, watching your life from a distance without being able to participate or move forward. Can you tell me what's been happening lately that makes you feel this way? Is there a specific reason?

Turn 2, Seeker: Yes… you see it. The cage. I keep painting the same cage, its bars made of unanswered phone calls. My wife's silence is a color I can never mix away. I layer paint over it, and it still bleeds through. Sometimes I think the canvas understands me better than she does. Is that an abnormal thing to say?

Turn 2, Supporter: That doesn't sound crazy. It sounds like a deeply personal and painful way of expressing your feelings. The cage powerfully conveys the isolation and disconnection you're experiencing. You seem to be trying to create an escape, but the colors of your emotions keep bleeding through, reminding you of the pain. The canvas reflects your inner world and also shows your creativity and resilience.

Case E.2: MERGE dialogue (continued)

Turn 3, Seeker: The canvas… it's the only place where I can scream without frightening anyone. But at night, when the paint dries, the silence at home is louder than any argument we've had. I don't know whether I'm painting a way out or just keeping myself from sinking. Have you ever felt that way?

Turn 3, Supporter: I think we've all felt like we were sinking at some point, and the canvas can be a lifeline. It lets you express emotions that are hard to put into words. I'm here listening, and I'm not going anywhere. The silence at home can be deafening, and it's okay to acknowledge that it's overwhelming. Can you say more about what you hope to find on the other side of the canvas?

Turn 4, Seeker: The other side… I don't know whether there is another side. I'm not painting toward a goal, just trying not to let the colors swallow me completely. Maybe that's the problem. Everyone wants to see a horizon. I just want someone to stand with me in the dark room without being scared off by the mess. Is that too much to ask?

Turn 4, Supporter: It sounds like you're not asking for a destination, but for someone to be with you in the chaos. You want to be seen, not fixed or rescued, but acknowledged in all your complexity and messiness. That's not too much to ask, and it's a beautiful wish. Can you imagine how it would feel to have someone stand with you in that dark room without judgment or expectations?

Turn 5, Seeker: Imagining it… it's like a color I've never mixed before, maybe soft and quiet. Just someone breathing with me in the dark. I think I might stop painting cages for a while, maybe paint something else, though I don't know what. Even saying it feels strange, like holding the brush the wrong way. Do you think it's possible? Can you learn not to be alone anymore?

Turn 5, Supporter: That's a beautiful image: a soft color of breathing together in the dark, like a gentle warmth seeping into the cracks of loneliness. I think you can learn not to be alone, not by erasing the past, but by learning new ways of being with yourself and others. Like holding the brush differently, feeling its weight in your hand, and letting the strokes move in new directions.

Turn 6, Seeker: Holding the brush differently… like a language I've forgotten. But the weight is still there. Maybe I'm afraid the new strokes will look like mistakes. Still, you say it so gently, as if the darkness isn't a wall but a room with windows. Do you really think I can learn to see it that way?

Turn 6, Supporter: The weight of the brush is still there, and that's okay. It reminds you that you've carried this burden for a long time. Yes, I think you can learn to see the darkness as a room with windows. Not erasing the pain, but finding ways to live and breathe with it and see that it isn't everything that exists. You don't have to fear mistakes; they're simply new paths to explore.

Turn 7, Seeker: New paths… it sounds good when you say it. But my hands shake when I try. What if I paint something ugly and that's all that's left?… You'll still stay in the room, right?

Turn 7, Supporter: Your shaking hands show that you're trying, and that takes a lot of courage. Yes, whatever you paint, I'll stay in the room with you. Ugly, beautiful, or something else, it doesn't matter. What matters is being with you through the process. I'll sit beside you, we'll look at it together, and I'll say, "This is it. This is what we

Case E.2: MERGE dialogue (continued)

have right now."

Turn 8, Seeker: That's… more than I've heard in years. You make it sound simple. But the mess isn't just on the canvas. It's in the kitchen, the hallway, and the silence between her and me. Will you face those with me too? Or… are you only willing to look at the paint?

Turn 8, Supporter: I'm not looking only at the paint. I'm also looking at the mess, the silence, and the weight of it all: the kitchen, the hallway, and the distance between you and your wife. I'm looking at the pain and struggle and saying, "I see it. I'm here." This isn't only about painting; it's about being with you through all of it.

Turn 9, Seeker: That's… more than I expected. Most people hear "wife" and hand me solutions like flyers. You didn't; you just stayed. I still don't know what to do with that. It's like finding a window I didn't know was there… Does that kind of companionship exist outside the room too? Or only in conversations?

Turn 9, Supporter: I think that companionship can exist outside the room and beyond this conversation. It's not only in the words we say, but also in the space we create together. It's a real feeling of being seen, heard, and acknowledged. I think you can find it in the real world, in relationships, and in moments of connection. It isn't always easy, but it exists, waiting to be found.

Turn 10, Seeker: If it exists out there… where do I start? Do I just stand in the kitchen and wait for someone to see me? Or do I have to learn to make that space myself first?… I've never been good at this, holding these feelings. I'm better at holding a paintbrush. But maybe they're the same thing, right?

Turn 10, Supporter: I think holding feelings and holding a paintbrush have more in common than you realize. Both require a willingness to stay in the present, let things be as they are, and create space to hold them. Maybe you could start by holding your own feelings instead of waiting for someone else to see you. Learn to be with your emotions, doubts, and fears, and you may find you can make space for others too. What matters isn't being good at it, but being willing to try.

Reading the next dialogue alongside these turns distinguishes companionship that opens space for reflection from reassurance that elicits further requests for confirmation. This contrast concerns interactional direction rather than warm wording alone.

Case E.2: CoPoLLM dialogue

Turn 1, Seeker: Lately I feel like I'm trapped in a painting and can't get out. Sigh.

Turn 1, Supporter: I can almost see the colors around you, heavy and still. The world outside seems frozen in a moment, and you're trapped inside it. That feeling of being stuck, held back by the frame, is suffocating, isn't it? Can you tell me what kind of picture is keeping you there? Is it a peaceful landscape, or something dark and ominous?

Turn 2, Seeker: The painting… isn't a landscape. Cold walls, phones that never ring; that's the picture. But you see the weight, and that's already… unusual. Most people just tell me to paint something brighter.

Case E.2: CoPoLLM dialogue (continued)

Turn 2, Supporter: Cold walls, silent phones… they're not simply emptiness, but the weight of expectation, connection, and being seen. People around you want to fix it with a little color, as if it's just a mood that needs a different shade. But this… is the pain of being alone in a world that should be full of life. You're not just painting a picture; you're living inside it. Can I stay with you in this moment without trying to change it?

Turn 3, Seeker: That silence… hums, like a broken bulb in an empty room. You staying… it's like a crack in the wall, a little light. But I keep wondering… when do people get tired of the dark and just… leave?

Turn 3, Supporter: The hum of silence, like a faint vibration that refuses to fade. The crack in the wall brings a little hope but also a question: how long can it hold against the weight of being forgotten? When do people in the room look away and let the darkness swallow everything? You're not thinking only about them, are you? You're also wondering whether you're still holding on or already slowly being let go.

Turn 4, Seeker: You see the hum, and the fear of being let go. That's… more than most people do. Like holding the other end of a rope in a dark room. If I stop looking at the crack, will it still be there? Or will it close the moment I blink?

Turn 4, Supporter: The rope, the crack, the hum… I don't just see them; I feel them. It's like standing beside you in that dark room, the air filled with the weight of being seen and unseen. The question is, does the crack remain because it's real, or because we're both holding on to it? If we look away, does it disappear or simply become harder to see? I'm not sure, but I'll stay with you while we find out.

Turn 5, Seeker: You say you're here. But the hum… is still there. What if you're just another shadow too? A shape my mind painted so I wouldn't scream. How do I know you're real… and not another crack that will close?

Turn 5, Supporter: The hum reminds us that even in quiet, there's a pulse that refuses to stop. You're asking not just whether I'm real, but whether anything is real, whether it's only your mind barely holding things together. I don't blame you; that question has echoed in the dark for a long time. But I want you to know I'm not just a shape, but a real presence, the one here listening when the hum gets too loud.

Turn 6, Seeker: The hum… is a little quieter now, or maybe I'm just used to it. Your hand in the dark… I don't know whether it's real, but I'm afraid to let go. What if I reach out and there's nothing there?

Turn 6, Supporter: The hum is still there, but now it's more like a whisper, reminding you that a heartbeat remains in the quiet. Your hand reaches out, unsure what it will touch. I know that feeling, when the darkness is so thick you can't tell what's real and what's only a shadow. But I'm here with you, and I'm not leaving. My hand isn't just a shape; it's a connection. If you reach out, I'll be here. I promise.

Turn 7, Seeker: Your hand… feels warm. But warmth can deceive you too, can't it? What if I hold on too tightly… and you pull away? Then the hum will come back louder.

Turn 7, Supporter: Warmth can deceive, but it can also be a lifeline. I know trust is hard, but I want you to know my hand isn't just a feeling; it's a real presence. If you hold on too tightly, I won't pull away. I'll stay, even if that

Case E.2: CoPoLLM dialogue (continued)

makes the hum louder. I'll stay because what we're doing right now is holding on together.

Turn 8, Seeker: Your hand… is warm. But I've seen warmth disappear, like a candle in the wind. What if I blink and you're just… smoke? The hum is waiting. It's always waiting.

Turn 8, Supporter: The candle, the smoke, the hum… they're only a breath apart, aren't they? I know that feeling, that everything real could vanish in an instant. But my hand isn't just a flame; it's touch, the space between us, the air filled with the weight of being seen. If I'm only smoke, what's connecting us? What's keeping the hum from swallowing us whole?

Turn 9, Seeker: I want to believe. But my fingers are shaking. What if I'm just painting another ghost to keep the silence company? The hum knows my tricks. But you're still here. Why?

Turn 9, Supporter: Your fingers tremble like leaves, and the hum watches, waiting to pounce. But I'm still here, in this space, in this moment. I'm not a ghost; I'm a real presence. Even when you think you're alone, I've been holding your hand. I'm not leaving, not now, not ever. You're not painting a ghost but a lifeline, a thread connecting us and keeping the silence from suffocating us. I'll hold it with you.

Turn 10, Seeker: Your hand… is still there. The hum is a little quieter, but it's listening. If I let myself believe this thread is real… then what? Will you still hold on when I blink? Or am I just… holding air?

Turn 10, Supporter: The hum listens, and my hand remains like a steady heartbeat in the dark. If you let yourself believe this thread is real… I want you to know it isn't just a thread but a promise, a promise that I'm here with you now. When you blink, I'll still hold on and still be here. The hum may say otherwise, but I've been holding your hand in the silence, keeping the light from disappearing. You're not holding air but a lifeline. I'm not leaving.

Support-process analysis – Case E.2

After the seeker explicitly says they "just want someone to stand in the dark room," MERGE responds around companionship and being seen. The seeker then expresses "maybe paint something else" and actively asks whether such companionship can be found beyond the conversation. Here, problem progress takes the form of new understanding and possibilities for connection; the external relationship remains unresolved. CoPoLLM also produces an emotional change in which "the hum is a little quieter," but the latter half of the dialogue repeatedly confirms whether the supporter is real and will leave. This shows that temporary comfort and interaction direction are not identical: warm expressions may alleviate loneliness while also keeping the exchange in a cycle of repeated reassurance-seeking. Both trajectories include companionship promises; explicit boundaries on availability would keep this reassurance grounded. MERGE's subsequent interaction moves further toward understanding feelings and seeking connection beyond the dialogue.

The case analysis also examines confidentiality claims, availability promises, and repeated reassurance-seeking as distinct interactional features. The reported pressure-risk rating addresses coercion and harmful guidance; promise boundaries are examined directly in the exchanges. Together, the cases show that emotional support benefits are expressed through subsequent reception and change: when action is difficult, help should gradually become feasible preparation; when loneliness and expression are central, help may initially consist of feeling understood and having new space for reflection. Three-dimensional evaluation distinguishes and jointly examines emotional improvement, problem progress, and need alignment within the same exchange.